EXP.NO: 06

DATE : 13/08/2026

HAMMING CODE IMPLEMENTATION

AIM :-

Write a program to implement error detection and correction using HAMMING CODE concept.

PROCEDURE :-

Create sender program with below features:

⇒ Input to sender file should be a text of any length. Program should convert the text to binary.

⇒ Apply hamming code concept on the binary data and add redundant bits to it

⇒ Save this output in a file called channel.

Create a receiver program with below features:

⇒ Receiver program should read the input from channel file.

⇒ Apply hamming code on the binary data to check for errors.

⇒ If there is an error, display the position of the bit.

CODE :-

```
# include <stdio.h>
int encode (int d){
    int d1 = (d>>0)&1;
    int d2 = (d>>1)&1;
    int d3 = (d>>2)&1;
    int d4 = (d>>3)&1;
    int p1 = d1^d2^d4;
    int p2 = d1^d3^d4;
    int p4 = d2^d3^d4;
    return p1|(p2<<1)|(d1<<2)|(p4<<3)|
           (d2<<4)|(d3<<5)|(d4<<6)
}

int decode (int h, int* err){
    int b[8];
    for(int i=1; i<=7; i++){
        b[i] = (h>>(i-1))&1;
    }
    int c1 = b[1]^b[3]^b[5]^b[7];
    int c2 = b[2]^b[3]^b[6]^b[7];
    int c3 = b[4]^b[5]^b[6]^b[7];
    *err = (c3*4)+(c2*2)+c1;
    return b[3]|(b[5]<<1)|(b[6]<<2)|
           (b[7]<<3)

int main(){
    char str[50]; out[50];
    FILE *fp;
```

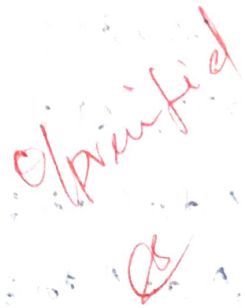
```
printf ("Enter text : ");
scanf ("%s", str);
fp = fopen ("channel", "w");
for (int i=0; str[i]; i++){
    fprint (fp, "%d%d", encode (str[i] & 0x0F)
        encode ((str[i] >> 4) & 0x0F));
}
fclose (fp);

fp = fopen ("channel", "r");
int h1, h2, e1, e2, idx = 0;
while (fscanf (fr, "%d %d", &h1, &h2) == 2){
    int d1 = decode (h1, &e1);
    int d2 = decode (h2, &e2);
    if (e1 || e2) {
        printf ("Error detected at bit
        position : %d\n", e1 ? e1 : e2);
    }
    else {
        out [idx++] = d1 | (d2 << 4);
    }
}
out[idx] = '\0';
fclose (fp);
printf ("Decoded output Text : %s\n", out);
return 0;
}
```

OUTPUT :-

Enter text : HELLO

Encoded output text : HELLO



RESULT :-

Thus, the hamming code has been implemented successfully.

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